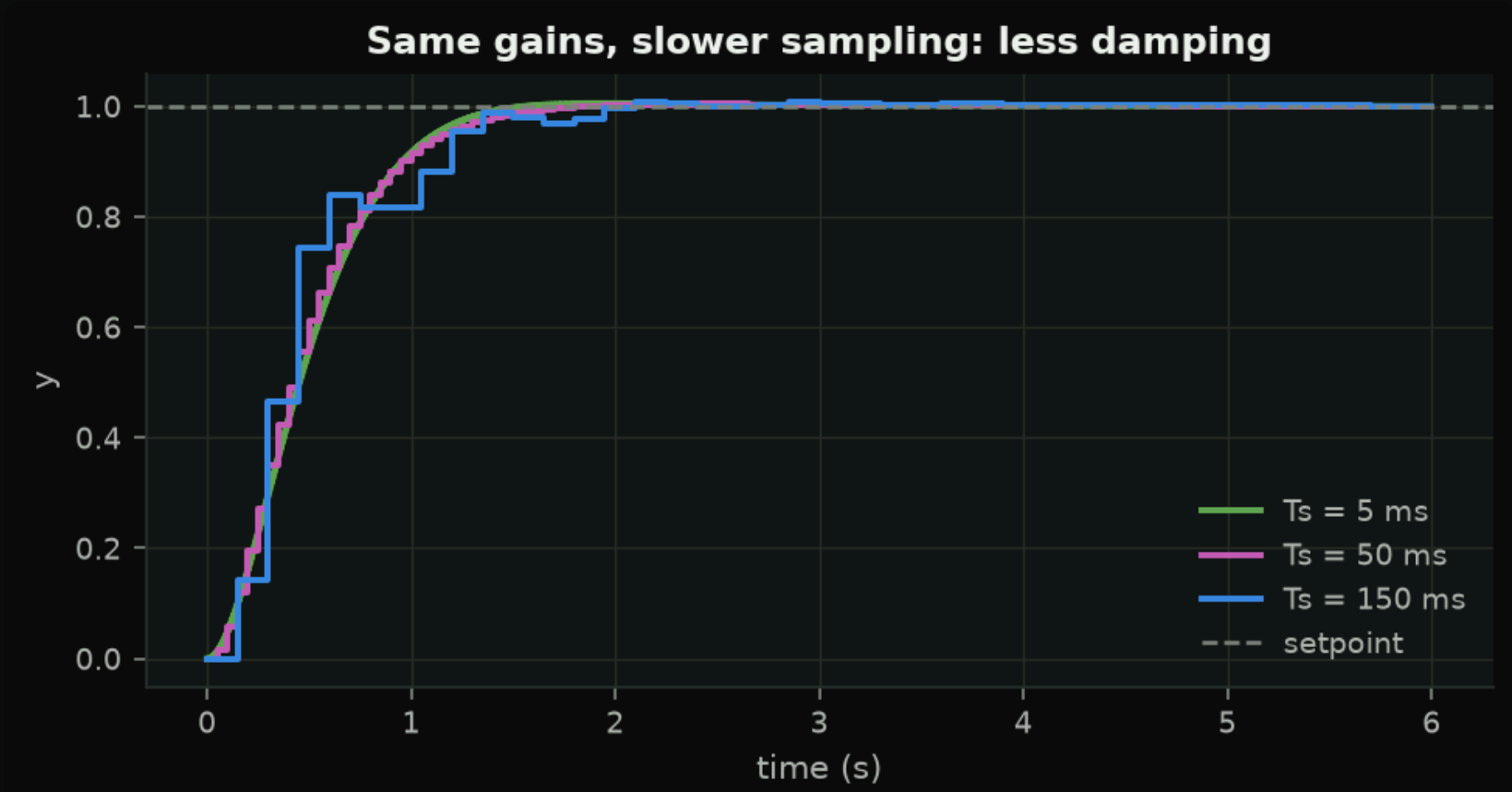
SC 02 · TUNING AND REAL-WORLD PID

Real hardware breaks textbook PID

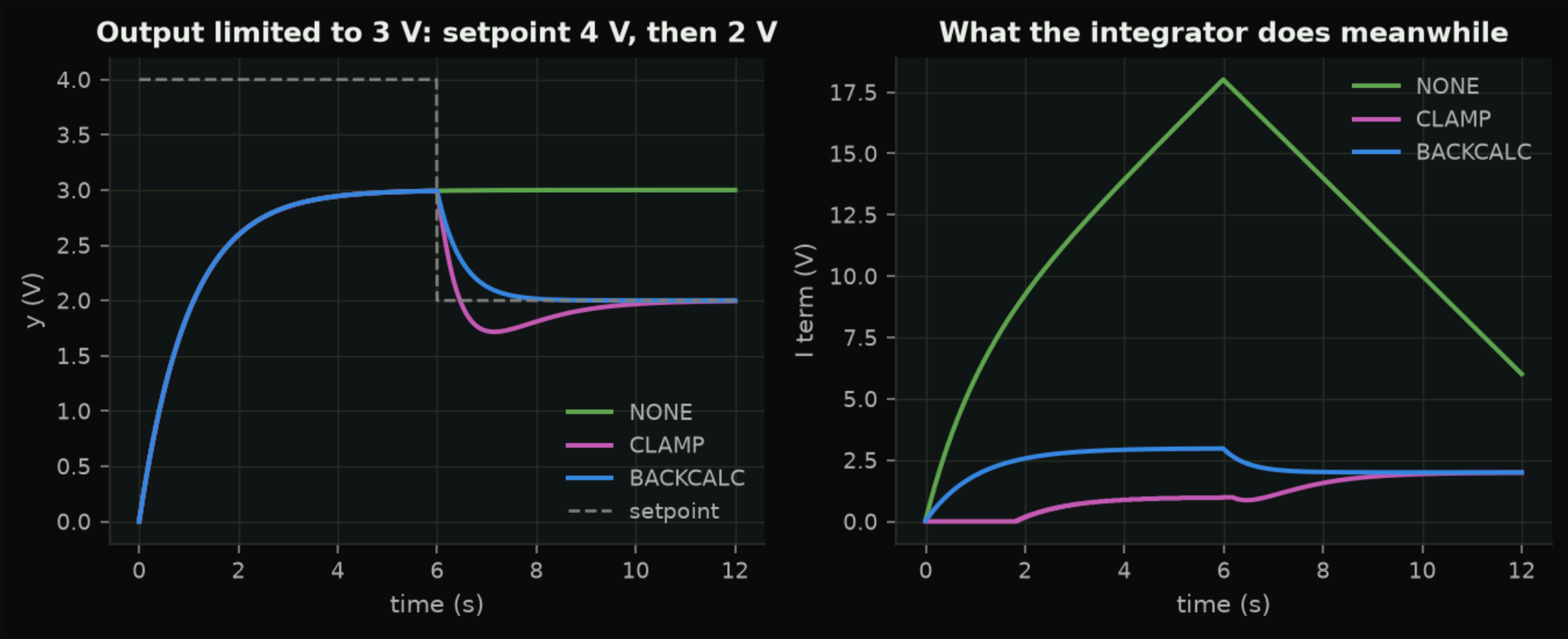
Five problems -- sampling, saturation, noise, setpoint jumps, unknown plants -- and the easyPID feature for each.

Sampling costs phase

$$y_{k+1} = e^{-T_s/\tau} y_k + K(1 - e^{-T_s/\tau}) u_k \qquad \text{hold delay} \approx T_s/2 \;\Rightarrow\; -360 f T_s/2 \text{ degrees}$$



Saturation and windup



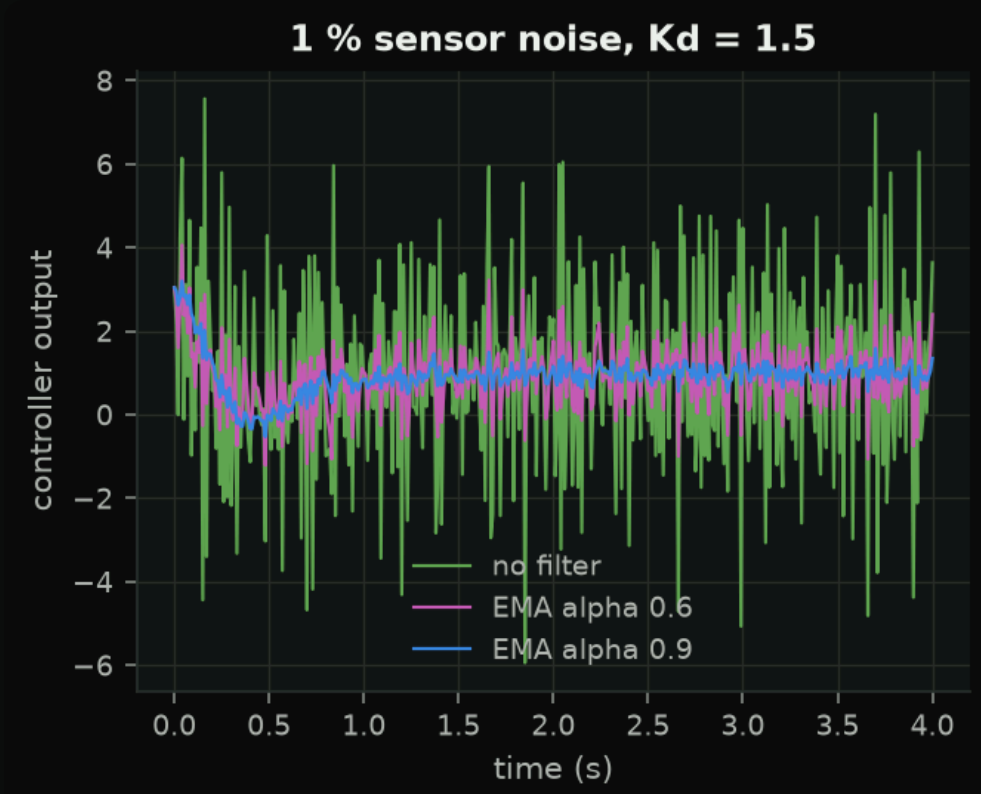
`ANTIWINDUP_CLAMP` (default): undo this sample's integration if it pushes further into the limit.

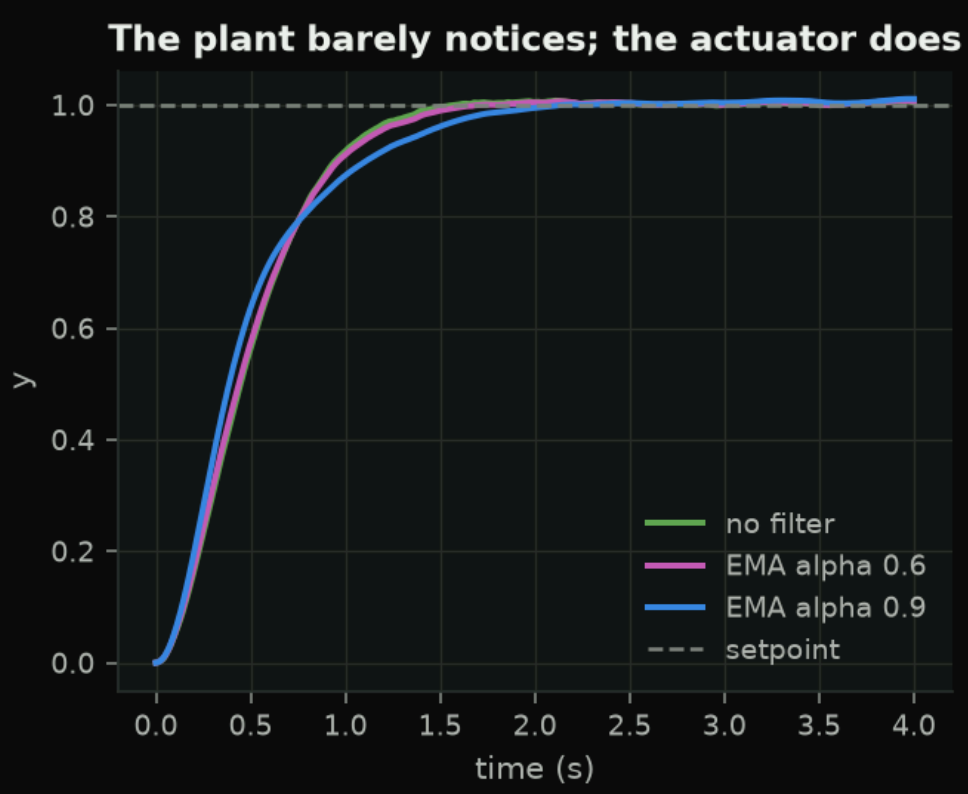
`BACKCALC` : bleed the excess back through $1/K_i$.

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Derivative noise: the EMA filter

$$d_k^f = \alpha d_{k-1}^f + (1-\alpha) d_k, \qquad f_c = -\frac{\ln \alpha}{2\pi T_s}$$





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Setpoint jumps kick the D term

easyPID differentiates the **error**: a setpoint step gives one sample of $K_d \Delta r / \Delta t$.

Fix: ramp the setpoint (`ramp 0.05` in the lab sketch), or differentiate the measurement.

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Measure, don't eyeball

metric	definition
rise time	10 % → 90 % of the step
overshoot	$100(y_{peak} - r)/(r - y_0)$
settling	stays within 2 % (or a ±band)
steady-state error	r − mean of the last 10 %

`bootcamp.metrics.step_info` -- used for every spec in the course.

Identify: first order plus dead time

$$G(s) = \frac{Ke^{-Ls}}{\tau s + 1}, \qquad \tau = 1.5(t_{63} - t_{28}), \quad L = t_{63} - \tau$$

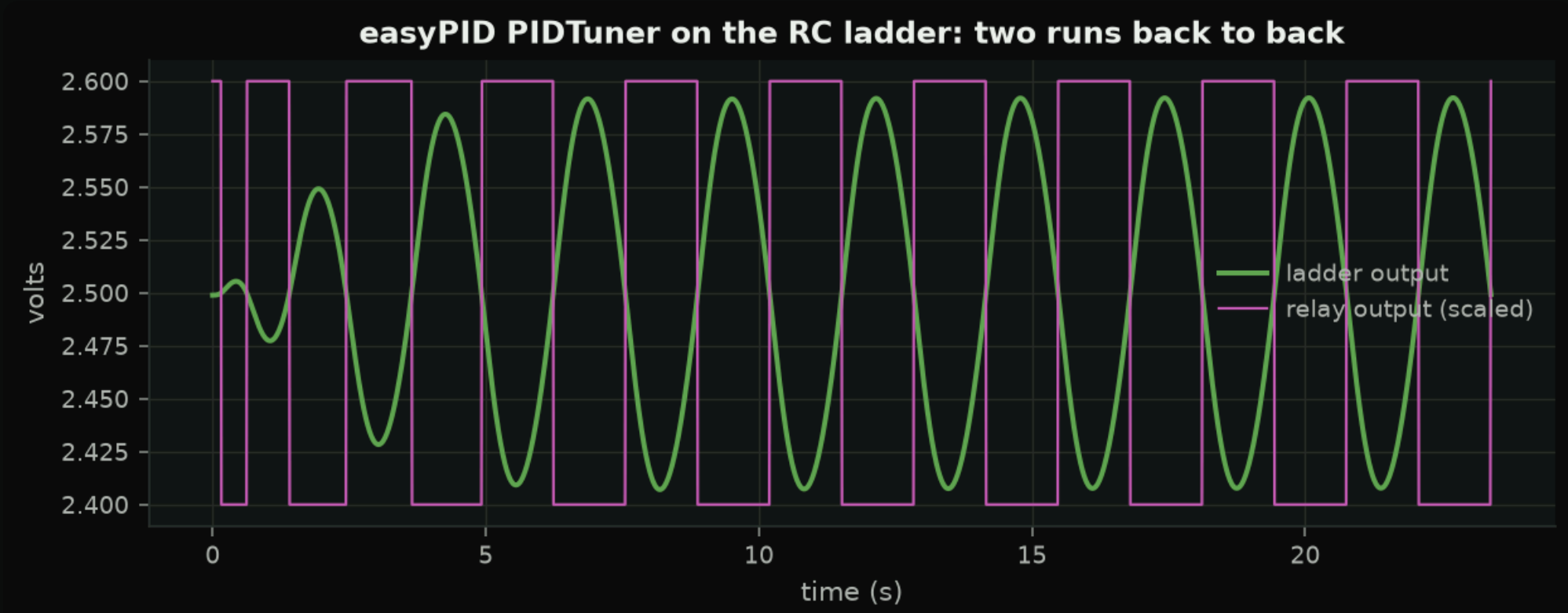
The 3-stage RC ladder (exact: $1/(1 + 6sT + 5s^2T^2 + s^3T^3)$) fits $K = 1, \tau \approx 5.1$ s, $L \approx 0.95$ s.

Reaction-curve rule (easyPID tuning guide): $K_p = 1.2\tau/(KL), K_i = K_p/2L, K_d = 0.5K_pL$.

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Relay autotuning: easyPID `PIDTuner`

$$K_u = \frac{4d}{\pi\sqrt{a^2 - h^2}}, \qquad P_u = \text{period}$$



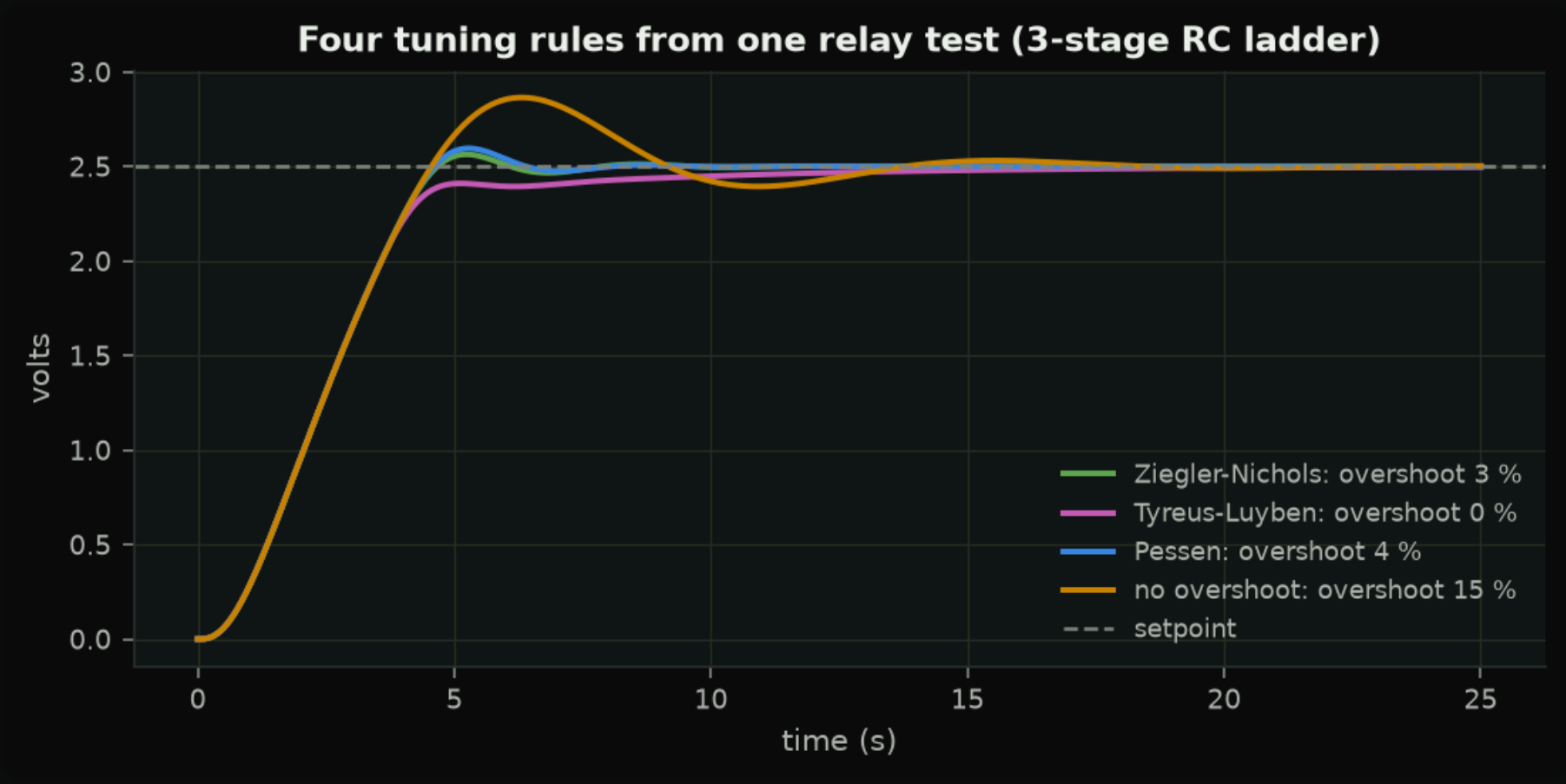
Test your tools on a known answer

RC ladder: exact $K_u = 29$, $P_u = 2.57$ s.

run	K_u	P_u
1st, from rest	42	2.04 s
2nd, straight after	28	2.64 s

The first run averages a still-growing oscillation. The lab sketch runs the tuner **twice**. The noise band (hysteresis) biases K_u low: keep h/a small.

Four rules from one test



Lab 2 · 3-stage RC ladder, `s02_rc_tuning`

- **A** open-loop step → FOPDT → reaction-curve gains
- **B** `tune 2.0 0.005` → K_u, P_u, four rules; check by hand
- **C** windup on purpose: `lim 0 3`, `aw none / clamp / backcalc`
- **D** `ts 200`, `filter 0.7` if time allows

Never relay-tune something that cannot oscillate safely -- like next week's rig.

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